

sTMS Interface Functional Specification

Temasek Polytechnic – CET Data Repository

Version 1.0

Document Version

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Document Acceptance Form

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1. Introduction

1.1 Purpose of this document

This Interface Functional Specification (IFS) is to provide the agreed baseline of communication requirements and detailed technical specifications between the SaaS shared Training Management System (sTMS) and Temasek Polytechnic (TP) CET data repository for transferring sTMS CET data, student photos and supporting documents.

sTMS shall identify all TP CET data with changes, generate corresponding Database File and CSV files for these data, deposit Database File, CSV files, student photos and supporting documents encrypted by PGP protocol in the SFTP server accordingly. New scheduled jobs to be implemented by Temasek Polytechnic will retrieve, decrypt and process these PGP encrypted data files accordingly for downstream systems' CET student data needs.

1.2 Intended Audience

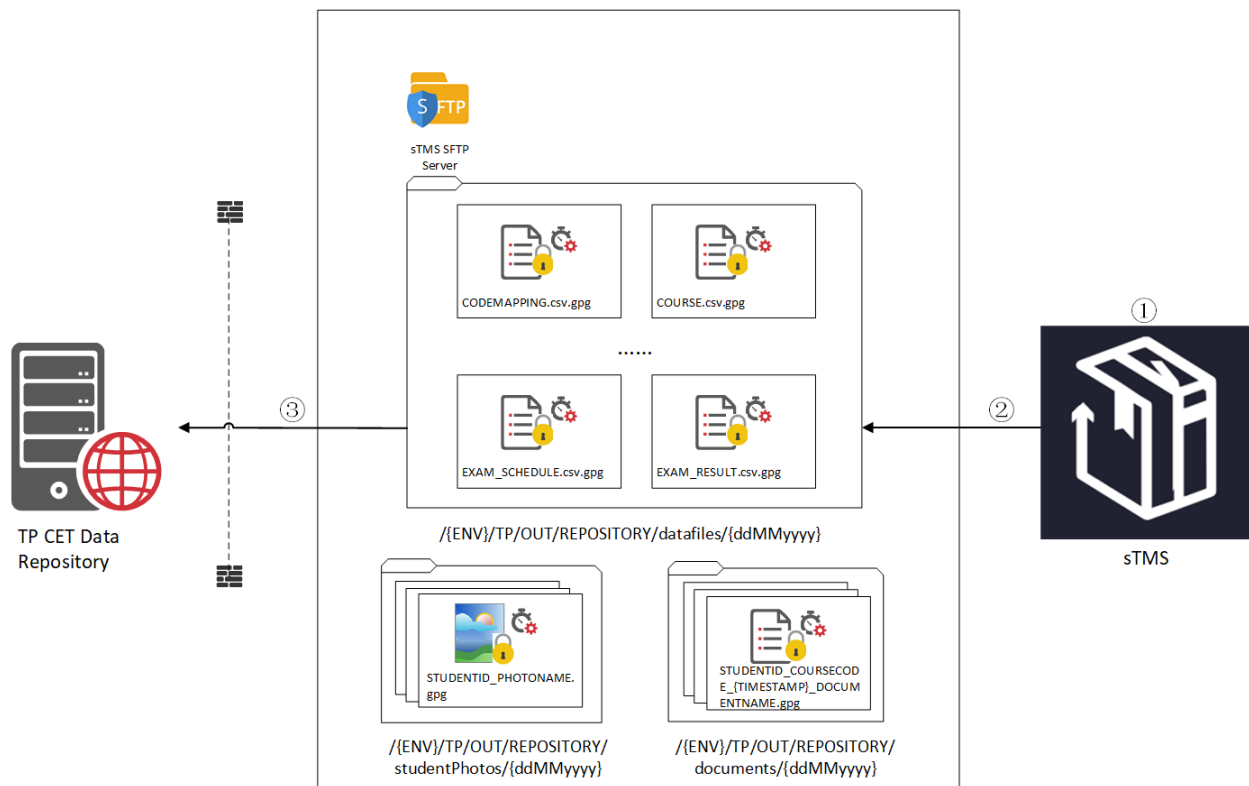
This technical document is used for system communication. Therefore, the intended audiences are the technical stakeholders of shared Training Management System and TP CET Data Repository System.

2. Technical Overview

2.1 System Flow

There are 3 roles involved in the interface diagram.

- **TP CET Data Repository:** Temasek Polytechnic CET data repository.
- **sTMS SFTP Server:** A SFTP server hosted in Azure that stores encrypted files used for data exchange.
- **sTMS:** shared Training Management System.



*The above system flow only shows some CSV files encryption with PGP protocol. For the interface file name of all TP CET data CSV files, please refer to [2.2 Interface Design Summary](#).

SN	Description
1	sTMS generates TP CET data CSV files based on institution, table name, modified time, and finds the corresponding student photos and supporting documents based on institution,

	student ID, modified time on a pre-defined schedule. For file generation rules can refer to 2.3. Business Rule
2	sTMS configures timer job to generate folders for different kind of resources: / {ENV} / TP / OUT / REPOSITORY / datafiles / {ddMMyyyy} / / {ENV} / TP / OUT / REPOSITORY / studentPhotos / {ddMMyyyy} / / {ENV} / TP / OUT / REPOSITORY / documents / {ddMMyyyy} / in the sTMS SFTP server daily. sTMS pushes TP CET data CSV files, student photos and supporting documents under {ddMMyyyy} folder after encryption with PGP protocol.
3	TP CET data repository pulls these CSV files, student photos and supporting documents from sTMS SFTP server on demand.

2.2 Interface Design Summary

Interface File Name	
	CODEMAPPING.CSV.gpg
	COURSE.CSV.gpg
	COURSERUN.CSV.gpg
	COURSERUNCLASSGROUPMEMBER.CSV.gpg
	COURSERUNENROLMENT.CSV.gpg
	COURSERUNENROLMENTINFO.CSV.gpg
	STUDENTINFORMATION.CSV.gpg
	STUDENTPERSONALINFO.CSV.gpg
	SUBJECT.CSV.gpg
	SCHOOL.CSV.gpg
	WORKEXPERENCE.CSV.gpg
	EDUCATIONALQUALIFICATION.CSV.gpg
	FACILITY.CSV.gpg
	SESSION.CSV.gpg
	SEMESTER.CSV.gpg
	STUDENTFINANCE.CSV.gpg
	STUDENTAWARD.CSV.gpg
	AWARDINFORMATION.CSV.gpg

	STUDENTFAMILYMEMBER.CSV.gpg EXAMSCHEDULE.CSV.gpg EXAMRESULT.CSV.gpg STUDENTID_PHOTONAME.gpg STUDENTID_COURSECODE_{TIMESTAMP}_DOCUMENTNAME.gpg
Data Source System	shared Training Management System
Data Destination System	Temasek Polytechnic CET Data Repository
Method	SFTP
File Path	/{ENV}/TP/OUT/REPOSITORY/datafiles/{ddMMyyyy}/ /{ENV}/TP/OUT/REPOSITORY/studentPhotos/{ddMMyyyy}/ /{ENV}/TP/OUT/REPOSITORY/documents/{ddMMyyyy}/ ENV: UAT, PROD
Encryption Method	PGP
Frequency	Daily Sync
Timing	The cut off time for changed data will be 12 am every day, timer job trigger timing is next day 01:00 am (TBC)
Folder Retention Policy	Folders under the /{ENV}/TP/OUT/REPOSITORY are retained for a maximum of 30 days in the sTMS SFTP server, after this period the folders will be automatically deleted. TP also can delete these folders after pulling out the files
File Transfer Policy	sTMS shall transfer empty files with header info into sTMS SFTP server if there are no new records are added/updated in the table
Possible Exception and Exception Handling	File loss or transfer failure due to sTMS internal exception. E.g., Timer service down, etc. Exception handling steps: • sTMS support team shall be informed by email of the transfer failure. • AP team checks the relevant logs to troubleshoot the cause of the failure. • If the issue is due to the transfer files could not be found in the newly created SFTP folder. ○ AP redeposits the files in the newly created corresponding SFTP folder.

2.3 Business Rule

CSV files generation rule

Table record trigger point: First-time full load data sync + Subsequent delta data sync.

Delta Data sync trigger point: Delta Data sync mode is record level. For example, if the area of interest field in the course table is updated, send all the data fields of this record in the course table to the TP CET data repository.

Filter condition:

- Institution = TP,
- The data fields in the table have changed,
- Modified datetime later than last successful sync datetime.

Besides TP only data, the global master data shared by all IHLs will be also included in the code mapping table. E.g., Citizenship Type, Course Category, Course Type, etc.

Student photos and supporting documents generation rule

Trigger point: Student upload/ update student photos or supporting documents.

Filter condition:

- Institution = TP,
- Student uploads/updates new student photo or supporting documents,
- Modified datetime later than last successful sync datetime,
- Users can identify student photo and support documents by the naming convention. E.g.,
STUDENTID_PHOTONAME.gpg,
STUDENTID_COURSECODE_{TIMESTAMP}_DOCUMENTNAME.gpg.
The Timestamp format is ddMMyyyyHHmmssSSS, where SSS in capital case represents the milliseconds.

3. Interface File Generation

3.1 CSV Files Interface Fields

3.1.1 CODEMAPPING Interface Fields

Field Name	Description	Data Type	Remarks
Action	C – Create; U – Update; D - Delete	String(\$nvarchar(1))	
Sourceld	Record unique Id	String(\$nvarchar(100))	A unique Id automatically generated by the database to ensure unique identification.
 Code	Code of code category	String(\$nvarchar(200))	Sample value: 1
 CodeCategory	Code category type	String(\$nvarchar(50))	Value list: CitizenshipType, CourseCategory, CourseType, NationalManpowerCluster, Branch, School, CountryCode, RaceCode
CodeValue	Code value of code category	String(\$nvarchar(500))	Sample value: Re-registered
Status	Code mapping status	String(\$nvarchar(20))	Value List: Active, Inactive

The symbol  represents the field is the primary key.

3.1.2 COURSE Interface Fields

Field Name	Description	Data Type	Remarks
Action	C – Create; U – Update; D - Delete	String(\$nvarchar(1))	
Sourceld	Record unique Id	String(\$nvarchar(100))	A unique Id automatically generated by the database to ensure unique identification.
 CourseCode	Course code	String(\$nvarchar(50))	Sample Value: TP-SD-FQ0001
 Version	Course version number	Number	Sample Value: 1
CourseName	Course name	String(\$nvarchar(200))	Allow input of multilingual and special characters (e.g., TM , ©, ®)
CourseShortName	Course short name	String(\$nvarchar(200))	Sample Value: SC1

CourseCategory	Course category code	String(\$nvarchar(200))	The different course categories offered by TP. Sample Value: 1
CourseType	Course type code	String(\$nvarchar(20))	Sample Value: 1
CtypeOSPF	Course type with value (Others/SGUS/SGUP/FullQual)	String(\$nvarchar(20))	Sample value: SGUP
DepartmentCode	School codes	String(\$nvarchar(20))	Sample value: 1;2
Level	Total number of course components	Int	If full qualification structure Course -> MC -> Module, Level = 3
Status	Course status	String(\$nvarchar(20))	Value list: Draft, Pending fee setup, Pending activation, Scheduled for activation, Active, Inactive.
CourseAlternativeName	Course alternative name	String(\$nvarchar(max))	Sample value: DBM
ApprovalDate	Course approval date	Datetime	Sample value: 2022-02-28T00:00:00.000
SunsetDate	Course sunset date given by TPG/MOE	Datetime	Sample value: 2023-02-28T00:00:00.000
AreaOfInterest	Course area of interest.	String(\$nvarchar(max))	Sample value: Accounting, Banking & Finance

The symbol  represents the field is the primary key.

3.1.3 COURSERUN Interface Fields

Field Name	Description	Data Type	Remarks
Action	C – Create; U – Update; D - Delete	String(\$nvarchar(1))	
SourceId	Record unique Id	String(\$nvarchar(100))	A unique Id automatically generated by the database to ensure unique identification.
 CourseRunNo	Course intake number	String(\$nvarchar(100))	Sample Value: TP-SD-FQ0001-220001
CourseCode	Course code	String(\$nvarchar(50))	Sample Value: TP-SD-FQ0001
 Version	Course version number	Number	Sample Value: 1
CourseLevel	Total number of course components	Int	If full qualification structure Course -> MC -> Module, Level = 3
FTPT	Full-time course or part-time course	String(\$nvarchar(20))	Value List: FT-Full time, PT-Part time

StartDate	Course intake start datetime	Datetime	Sample value: 2022-02-28T00:00:00.000
EndDate	Course intake end datetime	Datetime	Sample value: 2023-02-28T00:00:00.000
Status	Couse intake status	String(\$nvarchar(20))	Value list: Draft, Pending fee setup, Ready for application, Open for application, Application closed, Confirmed, Commenced.
Semester	Course intake of which semester Value: AcademicYear + SemesterTerm	String(\$nvarchar(32))	Sample Value: 2022-S1

The symbol  represents the field is the primary key.

3.1.4 COURSERUNCLASSGROUPMEMBER Interface Fields

This table identifies what class the student/instructor is in and the class under which course intake, pathway, module.

If a student takes an FQ course, and selected two pathways under the course, each pathway selected 2 modules, then this table will contain 4 records for the student.

Field Name	Description	Data Type	Remarks
Action	C – Create; U – Update; D - Delete	String(\$nvarchar(1))	
Sourceld	Record unique Id	String(\$nvarchar(100))	A unique Id automatically generated by the database to ensure unique identification.
SubjectCode	Code of the module	String(\$nvarchar(50))	Sample Value: BM0001
SubjectName	Name of the module	String(\$nvarchar(200))	Sample Value: Business management
CoursePathwayId	Course pathway Id	String(\$nvarchar(50))	
FqCourseCode	Course code	String(\$nvarchar(50))	Sample Value: TP-SD-FQ0001
FqCourseRunNo	Course intake number	String(\$nvarchar(100))	Sample Value: TP-SD-FQ0001-220001
ComponentType	Class type	String(\$nvarchar(20))	Sample Value: Lecture
GroupNo	Class name	String(\$nvarchar(50))	Sample Value: Class1
 MemberId	Student or Lecturer Id	String(\$nvarchar(50))	Student – UAM Common ID Lecturer – Workday ID
MemberType	Identification of student or lecturer	String(\$nvarchar(50))	Value List: Student, Lecturer

SubjectEnrolmentStatus	Module enrolment status	String(\$nvarchar(50))	Values List: Not started, In-progress, Finished, Make-up exam, Deferred assessment, Drop, Quit, Granted Exemption, Granted Credit Transfer
FqEnrolmentStatus	Full qualification course type enrolment status sTMS would only send datafield if it is a full qualification course.	String(\$nvarchar(50))	Sample Value: Enrolled, Withdrawn, Pending for Enrolment
 ClassId	Class unique id	String(\$nvarchar(100))	

The symbol  represents the field is the primary key.

3.1.5 COURSERUNENROLMENT Interface Fields

Field Name	Description	Data Type	Remarks
Action	C – Create; U – Update; D - Delete	String(\$nvarchar(1))	
Sourceld	Record unique Id	String(\$nvarchar(100))	A unique Id automatically generated by the database to ensure unique identification.
 EnrolmentId	Enrolment Id	String(\$nvarchar(100))	Sample Value: Unique Id
StudentId	UAM Common Id	String(\$nvarchar(11))	
CourseRunNo	Course intake number	String(\$nvarchar(100))	Sample Value: TP-SD-FQ0001-220001
SubjectCode	CourseLevel is 1, this field will send Module code, CourseLevel is 3, SubjectCode is Null	String(\$nvarchar(50))	Sample Value: BM0001
FTPT	Full-time course or part-time course	String(\$nvarchar(2))	Value List: FT-Full time, PT-Part time
CourseCode	Course code	String(\$nvarchar(50))	Sample Value: TP-SD-FQ0001
CourseLevel	Total number of course components	Int	If full qualification structure Course -> MC -> Module, Level = 3. Use Level = 1 to indicate the subject/module
EnrolmentCourseLevel	1=subject, 2=PDC/MC/Course, 3=course Full Qualification = 3 if not Full Qualification = 2	Int	sTMS will only send 2 or 3.
EnrolmentStatus	Defines the student course intake level enrolment status of eligibility criteria.	String(\$nvarchar(50))	Sample Value: Enrolled, Deferred

EnrolmentDate	Enrolment Date	Datetime	Sample value: 2022-02-28T00:00:00.000
StatusUpdateDate	Student enrolment status modified date	Datetime	Sample value: 2022-02-28T00:00:00.000

The symbol  represents the field is the primary key.

3.1.6 COURSERUNENROLMENTINFO Interface Fields

Field Name	Description	Data Type	Remarks
Action	C – Create; U – Update; D - Delete	String(\$nvarchar(1))	
Sourceld	Record unique Id	String(\$nvarchar(100))	A unique Id automatically generated by the database to ensure unique identification.
 EnrolmentId	Enrolment Id	String(\$nvarchar(100))	Sample Value: Unique Id
CGPA	Cumulative grade point average for graduation	Number (2 decimal)	Sample Value: 3.90

The symbol  represents the field is the primary key.

3.1.7 STUDENTINFORMATION Interface Fields

Student information is single model table, if a student takes multiple courses, follow the below sorting logic to generate the record.

1. Enrolment status (Enrolled > others),
2. MOE course indicator (MOE Courses > Non-MOE Courses)
3. Course start date and end date (Follow semester/term start date and end date > Specify start date and end date),
4. Course duration (longer > shorter),
5. Course end date (latest > earliest).

Field Name	Description	Data Type	Remarks
Action	C – Create; U – Update; D - Delete	String(\$nvarchar(1))	
Sourceld	Record unique Id	String(\$nvarchar(100))	A unique Id automatically generated by the database to ensure unique identification.

 StudentId	UAM Common Id	String(\$nvarchar(11))	
Name	Full name	String(\$nvarchar(200))	Sample Value: Jake Tan
TPemail	Student school email	String(\$nvarchar(100))	NIL
PersonalEmail	Student personal email	String(\$nvarchar(100))	Sample Value: Jake@email.com
Firstname	NA	String(\$nvarchar(100))	NIL
Lastname	NA	String(\$nvarchar(100))	NIL
NRICFIN	NRIC/FIN number	String(\$nvarchar(20))	Sample Value: G1234567P
Passport	Passport number	String(\$nvarchar(20))	Sample Value: E12345678
HomeNo	NA	String(\$nvarchar(20))	NIL
Birthday	Date of birth	Datetime	Sample value: 2002-02-28T00:00:00.000
MobileNo	Mobile number	String(\$nvarchar(20))	Sample value: 12345678
Gender	Student gender	String(\$nvarchar(20))	Value list: Male, Female
Nationality	Nationality	String(\$nvarchar(20))	Sample Value: Singapore
Citizenship	Citizenship type	String(\$nvarchar(20))	Value List: Singaporean Citizen, Singapore PR, Foreigner
Race	Race	String(\$nvarchar(20))	Sample Value: Chinese
CountryOfBirth	Country of birth	String(\$nvarchar(20))	Sample Value: Singapore
CourseCode	Course code	String(\$nvarchar(100))	Sample Value: TP-SD-FQ0001
CourseName	Course name	String(\$nvarchar(200))	Sample Value: Short course 1
CourseShortName	Course short name	String(\$nvarchar(200))	Sample Value: SC1
 CourseIntakeNo	Course intake number	String(\$nvarchar(100))	Sample Value: TP-SD-FQ0001-220001
CourseLevel	Course level	Int	Sample Value: 3
DepartmentCode	School code	String(\$nvarchar(32))	
CGPA	Cumulated GPA	Number (2 decimal)	

EnrolmentStatus	Defines the student course intake level enrolment status of eligibility criteria.	String(\$nvarchar(50))	Sample Value: Enrolled, Deferred
EnrolmentStatusUpdateDate	NA	-	NIL
CourseExpiryDate	Sunset Date	Datetime	Sample value: 2023-02-28T00:00:00.000
 EnrolmentDate	Date of enrolment	Datetime	Sample value: 2022-02-28T00:00:00.000
CourseType	Course type code	String(\$nvarchar(20))	Sample Value: 1
CtypeOSPF	Course type with value of Others/SGUS/SGUP/FullQual	String(\$nvarchar(20))	SGUS
CourseCategory	Course category code	String(\$nvarchar(200))	Sample Value: 1
AddressIsFormatted	Yes or no to indicate Singapore Address	String(\$nvarchar(1000))	Value List: Yes, No
Block	Block or building number	String(\$nvarchar(max))	Sample value: BLK123
Street	Street name	String(\$nvarchar(max))	Sample value: Street A
UnitNo	Unit number	String(\$nvarchar(10))	Sample value: 33
FloorNo	Floor number	String(\$nvarchar(10))	Sample value: 08
BuildingName	Building name	String(\$nvarchar(max))	Sample value: Building A
PostalCode	Postal code	String(\$nvarchar(20))	Sample value: 307717
City	Country	String(\$nvarchar(100))	Sample value: Singapore
CountryCode	Country Code	String(\$nvarchar(6))	Sample value: SG
AddressLine1	Store address in lines of texts when address is not structured.	String(\$nvarchar(500))	
AddressLine2	Store address in lines of texts when address is not structured.	String(\$nvarchar(500))	
AddressLine3	Store address in lines of texts when address is not structured.	String(\$nvarchar(500))	

The symbol  represents the field is the primary key.

3.1.8 STUDENTPERSONALINFO Interface Fields

Field Name	Description	Data Type	Remarks
Action	C – Create; U – Update; D - Delete	String(\$nvarchar(1))	
SourceId	Record unique Id	String(\$nvarchar(100))	A unique Id automatically generated by the database to ensure unique identification.
 StudentId	UAM Common Id	String(\$nvarchar(11))	
Name	Full name	String(\$nvarchar(200))	Sample Value: Jake Tan
TPemail	Student school email	String(\$nvarchar(100))	NIL
PersonalEmail	Student personal email	String(\$nvarchar(100))	Sample Value: Jake@email.com
Firstname	NA	String(\$nvarchar(100))	NIL
Lastname	NA	String(\$nvarchar(100))	NIL
NRICFIN	NRIC/FIN number	String(\$nvarchar(20))	Sample Value: G1234567P
Passport	Passport number	String(\$nvarchar(20))	Sample Value: E12345678
HomeNo	NA	String(\$nvarchar(20))	NIL
Birthday	Date of Birth	Datetime	Sample value: 2002-01-01T00:00:00.000
MobileNo	Mobile number	String(\$nvarchar(20))	Sample value: 12345678
Gender	Student gender	String(\$nvarchar(20))	Value list: Male, Female
Nationality	Nationality	String(\$nvarchar(20))	Sample Value: Singapore
Citizenship	Citizenship type	String(\$nvarchar(20))	Value List: Singaporean Citizen, Singapore PR, Foreigner
Race	Race	String(\$nvarchar(20))	Sample Value: Chinese
CountryOfBirth	Country of birth	String(\$nvarchar(20))	Sample Value: Singapore
AddressIsFormatted	Yes or no to indicate Singapore Address	String(\$nvarchar(1000))	Value List: Yes, No
Block	Block or building number	String(\$nvarchar(max))	Sample value: BLK123
Street	Street name	String(\$nvarchar(max))	Sample value: Street A

UnitNo	Unit number	String(\$nvarchar(10))	Sample value: 33
FloorNo	Floor number	String(\$nvarchar(10))	Sample value: 08
BuildingName	Building name	String(\$nvarchar(max))	Sample value: Building A
PostalCode	Postal code	String(\$nvarchar(20))	Sample value: 307717
City	Country	String(\$nvarchar(100))	Sample value: Singapore
CountryCode	Country Code	String(\$nvarchar(6))	Sample value: SG
AddressLine1	Store address in lines of texts when address is not structured.	String(\$nvarchar(500))	
AddressLine2	Store address in lines of texts when address is not structured.	String(\$nvarchar(500))	
AddressLine3	Store address in lines of texts when address is not structured.	String(\$nvarchar(500))	

The symbol  represents the field is the primary key.

3.1.9 SUBJECT Interface Fields

Field Name	Description	Data Type	Remarks
Action	C – Create; U – Update; D - Delete	String(\$nvarchar(1))	
Sourceld	Record unique Id	String(\$nvarchar(100))	A unique Id automatically generated by the database to ensure unique identification.
 SubjectCode	Module code	String(\$nvarchar(50))	Sample Value: BM0001
 Version	Module version number	Number	Sample Value: 1
SubjectName	Module name	String(\$nvarchar(200))	Sample Value: Business management
SubjectShortName	NA	-	Not found in sTMS Default t null
DepartmentCode	School code	String(\$nvarchar(100))	Sample Value: BUS
Status	Module status	String(\$nvarchar(20))	Value list: Draft, Pending activation, Scheduled for activation, Active, Inactive
EffectiveFrom	Module effective from	Datetime	Sample Value: 2021-02-28T00:00:00.000
EffectiveTo	Module effective to	Datetime	Sample Value: 2022-02-28T00:00:00.000

Certificates	Module certificates name	String(\$nvarchar(max))	Sample Value: Cert1;Cert2
Component	Module component	String(\$nvarchar(max))	Sample Value: Lecture, Tutorial
NumberOfModule Hours	Module number of hours	Number (2 decimal)	Sample Value: 10.00
Examinable	Module is examinable	String(\$nvarchar(1))	Value list: Y – Yes; N - No
IncludedInGrading Basis	Module included in grading basis/GPA	String(\$nvarchar(1))	Value list: Y – Yes; N - No

The symbol  represents the field is the primary key.

3.1.10 SCHOOL Interface Fields

The academic schools under the IHL or branch of the IHL.

Below are the data fields for school:

Field Name	Description	Data Type	Remarks
Action	C – Create; U – Update; D - Delete	String(\$nvarchar(1))	
SourceId	Record unique Id	String(\$nvarchar(100))	A unique Id automatically generated by the database to ensure unique identification.
 SchoolCode	School code	String(\$nvarchar(32))	Sample Value: BUS
SchoolName	Name of the school	String(\$nvarchar(100))	Sample Value: School of Business
CreatedTime	School info creation time	Datetime	Sample Value: 2021-01-28T02:01:01.123
ModifiedOn	School info last modified time	Datetime	Sample Value: 2022-02-28T00:00:00.000
LastUpdatedBy	School info modified by	String(\$nvarchar(100))	Sample Value: Jake

The symbol  represents the field is the primary key.

3.1.11 WORKEXPERIENCE Interface Fields

Student will be able to view all information related to the student including personal information, contact information and academic information such as qualification and work experience.

Below are the data fields for work experience:

Field Name	Description	Data Type	Remarks
Action	C – Create; U – Update; D - Delete	String(\$nvarchar(1))	
SourceId	Record unique Id	String(\$nvarchar(100))	A unique Id automatically generated by the database to ensure unique identification
StudentId	UAM Common Id	String(\$nvarchar(11))	
 WorkExperienceId	Student work experience record id	String(\$nvarchar(100))	Format: uuid
DateFrom	Start date of relevant work experience	Datetime	Sample Value: 2021-02-28T00:00:00.000
DateTo	End date of relevant work experience	Datetime	Sample Value: 2022-02-28T00:00:00.000
Duration	Duration of the relevant work experience.	Int	The unit is month Sample value: 12
WorkExperienceType	Type of the relevant work experience.	String(\$nvarchar(32))	Value List: FT-Full time, PT-Part time
FieldofWork	Field of work of relevant work experience	String(\$nvarchar(max))	Sample Value: IT
NameOfEmployer	The employer's name for the relevant work experience.	String(\$nvarchar(max))	Sample Value: Avepoint
PositionHeld	The position held for the relevant work experience.	String(\$nvarchar(max))	Sample Value: Associates
BriefJobDescription	Brief description for work	String(\$nvarchar(max))	Sample Value: Data analyst

The symbol  represents the field is the primary key.

3.1.12 EDUCATIONALQUALIFICATION Interface Fields

Student will be able to view all information related to the student including personal information, contact information and academic information such as qualification and work experience.

Below are the data fields for educational qualification:

Field Name	Description	Data Type	Remarks
Action	C – Create; U – Update; D - Delete	String(\$nvarchar(1))	
SourceId	Record unique Id	String(\$nvarchar(100))	A unique Id automatically generated by the database to ensure unique identification

StudentId	UAM Common Id	String(\$nvarchar(11))	
 EducationId	Student education record id	String(\$nvarchar(100))	Format: uuid
QualificationType	Qualifiction Type	String(\$nvarchar(max))	Sample Value: No Formal Qualification, Pre-Primary, Lower Primary, Primary, Lower Secondary
FTPT	Full-time course or part-time course	String(\$nvarchar(20))	Value List: FT-Full time, PT-Part time
ExamCert	Exam or Certificate of qualification	String(\$nvarchar(max))	Sample Value: GCE NA LEVEL, GCE NT LEVEL, GCE O LEVEL
Grade	Grade of the qualifications	Number(2 decimal)	Sample value: 4.00
YearOfAward	Year of award for the certification	Datetime	Sample value: 2023-02-28T00:00:00.000
AwardInstitution	The award institution of the qualification	String(\$nvarchar(max))	Sample value: TP

The symbol  represents the field is the primary key.

3.1.13 FACILITY Interface Fields

Facility info will help both CET user and student to identify the location of the facility that has been assigned for a session/exam.

Below are the data fields for facility:

Field Name	Description	Data Type	Remarks
Action	C – Create; U – Update; D - Delete	String(\$nvarchar(1))	
SourceId	Record unique Id	String(\$nvarchar(100))	A unique Id automatically generated by the database to ensure unique identification
 FacilityId	Facility Id	String(\$nvarchar(32))	Sample Value: F0111
FacilityName	Facility name	String(\$nvarchar(100))	Sample Value: Facility A
Room	Room name of the facility	String(\$nvarchar(max))	Sample value: SOB room 1
UnitNo	Unit number of the facility	String(\$nvarchar(10))	Sample value: 33
BlockNo	Block or building number of the facility	String(\$nvarchar(max))	Sample value: BLK123
FloorNo	Floor number of the facility	String(\$nvarchar(10))	Sample value: 08

StreetName	Street name of the facility	String(\$nvarchar(max))	Sample value: Street A
BuildingName	Building name of the facility	String(\$nvarchar(max))	Sample value: Building A
PostalCode	Postal code of the facility	String(\$nvarchar(20))	Sample value: 307717
WheelChairAccess	Wheelchair accessibility option	String(\$nvarchar(1))	Value list: Y – Yes, N - No
Capacity	Capacity of the facility	String(\$nvarchar(20))	Sample value: 10-20
Equipment	Equipment of the facility	String(\$nvarchar(max))	Sample value: Monitor; Projector
Remarks	Remarks	String(\$nvarchar(max))	

The symbol  represents the field is the primary key.

3.1.14 SESSION Interface Fields

For timetabling setup process, sTMS provide an end-to-end solution to allow CET user to create session, configure session and publish them to the related students in student portal.

Below are the data fields for session:

Field Name	Description	Data Type	Remarks
Action	C – Create; U – Update; D - Delete	String(\$nvarchar(1))	
SourceId	Record unique Id	String(\$nvarchar(100))	A unique Id automatically generated by the database to ensure unique identification
Name	Session name	String(\$nvarchar(max))	Sample value: Session1
 SessionId	Session record Id	String(\$nvarchar(100))	Format: uuid
ClassId	Class unique id	String(\$nvarchar(100))	Format: uuid
CourseCode	Course code	String(\$nvarchar(50))	Sample Value: TP-SD-FQ0001
CourseIntakeNo	Course intake number	String(\$nvarchar(100))	Sample Value: TP-SD-FQ0001-220001
ModuleCode	Code of the module	String(\$nvarchar(50))	Sample Value: BM0001
GroupNo	Class Name	String(\$nvarchar(32))	Sample value: Class1
Topic	Session topic	String(\$nvarchar(max))	Sample value: Business Requirement Gathering

RecordAttendance	Define if the session will record attendance	String(\$nvarchar(10))	Value list: Yes, No
StartTime	Session start time	Datetime	Sample value: 2023-02-01T09:00:00.000
EndTime	Session end time	Datetime	Sample value: 2023-02-01T11:00:00.000
ModeOfTraining	Mode of Training of the session	String(\$nvarchar(64))	Value list: Classroom, Asynchronous eLearning, In-house
FacilityId	Facility Id	String(\$nvarchar(32))	Sample Value: F0111
Duration	training hours of the trainer.	Number (2 decimal)	Unit is hour Sample value: 2.00
Link	Link URL of the learning material of the session	String(\$nvarchar(max))	

The symbol  represents the field is the primary key.

3.1.15 SEMESTER Interface Fields

Semester/Term defines the standard semester/term start date and end date for usage in course intake setup.

Below are the data fields for semester:

Field Name	Description	Data Type	Remarks
Action	C – Create; U – Update; D - Delete	String(\$nvarchar(1))	
SourceId	Record unique Id	String(\$nvarchar(100))	A unique Id automatically generated by the database to ensure unique identification
AcademicYear	Academic year of semester/ term	String(\$nvarchar(max))	Sample Value: 2022
SemesterTerm	The value of semester/ term	String(\$nvarchar(max))	Sample Value: S1
 Semester	Course intake of which semester Value: AcademicYear + SemesterTerm	String(\$nvarchar(32))	Sample Value: 2022-S1
StartDate	The start date of the semester/term	Datetime	Sample value: 2023-02-28T00:00:00.000
EndDate	The end date of the semester/term	Datetime	Sample value: 2023-08-28T00:00:00.000

The symbol  represents the field is the primary key.

3.1.16 STUDENTFINANCE Interface Fields

Payment collection is located under the Finance process, and it is a centralized place for users with billing manager roles to manage all the payment collected in system or off the system.

Payment collection record includes the type of payment mode performed for course fee, student service fee, disciplinary fines and q2 evaluation fee. Depending on the chosen payment mode by applicant/company during payment summary page, payment collection record will be auto created by the system.

Below are the data fields for payment collection:

Field Name	Description	Data Type	Remarks
Action	C – Create; U – Update; D - Delete	String(\$nvarchar(1))	
SourceId	Record unique Id	String(\$nvarchar(100))	A unique Id automatically generated by the database to ensure unique identification
 PaymentCollectionId	Payment collection Id	String(\$nvarchar(20))	Sample Value: PC0001
ChargeFor	The type of billing charge for	String(\$nvarchar(20))	Value List: Course, Q2 evaluation, Student service, Disciplinary fine, Supplementary fee.
BillingReferenceNo	Billing reference number	String(\$nvarchar(100))	Sample Value: TPBilling1220210001
InvoiceNo	Invoice number	String(\$nvarchar(100))	Sample Value: TPINV0320220001
StudentId	UAM Common Id	String(\$nvarchar(11))	
DebtorType	Billing debtor type	String(\$nvarchar(10))	Value List: Company, Individual
NRICFIN	Student NRIC/FIN No. Displayed when the debtor type is individual	String(\$nvarchar(10))	Sample Value: S1234567E
UEN	Company UEN. Displayed when the debtor type is company	String(\$nvarchar(10))	Sample Value: OR220312
PaymentMode	Payment mode	String(\$nvarchar(100))	Value list: PSEA (Adhoc withdrawal form), PSEA (Standing order form), SkillsFuture Credits, Credit Card (e-payment), Debit Card (e-payment), eNETS (e-payment), PayNow (e-payment), AXS, GIRO, Counter Payment, Cheque, NETS, Telegraphic transfer

PaymentAmount	Payment amount, indicate amount made for the type of payment mode selected	Number (2 decimal)	Sample Value: 168.00
PaymentStatus	Payment status	String(\$nvarchar(20))	Value list: Refer to payment status table
PaymentDate	Payment date	Datetime	Sample value: 2023-02-28T00:00:00.000
ReceiptNo	Receipt number	String(\$nvarchar(100))	Sample Value: TPINV0320220001
ReceivedPaymentFrom	Received payment from	String(\$nvarchar(20))	Sample Value: MOE
Remarks	Payment collection remarks	String(\$nvarchar(max))	
FormSG	Displayed when payment mode is PSEA Adhoc/SO	String(\$nvarchar(1))	Value list: Y, N
BatchNo	Displayed when payment mode is PSEA Adhoc/SO	String(\$nvarchar(100))	Sample Value: 2845WL202112U10078
BankCode	Bank code Displayed when payment mode is GIRO	String(\$nvarchar(100))	Sample Value: 1234567
SwiftBICCode	Swift BIC code Displayed when payment mode is GIRO	String(\$nvarchar(100))	Sample Value: 4083156
BankAcctNo	Bank account code Displayed when payment mode is GIRO	String(\$nvarchar(100))	Sample Value: 254411634251
AcctHolderName	Bank account holder code Displayed when payment mode is GIRO	String(\$nvarchar(100))	Sample Value: Jake
AcctHolderContact	Bank account holder contact Displayed when payment mode is GIRO	String(\$nvarchar(20))	Sample Value: 78901234
AcctHilderEmail	Bank account holder email address Displayed when payment mode is GIRO	String(\$nvarchar(100))	Sample Value: jake@email.com
ClaimId	SFC claim Id Displayed when payment mode is SFC	String(\$nvarchar(20))	Sample Value: 2000217512
SFCPayoutRequestId	SFC Payout request Id Displayed when payment mode is SFC	String(\$nvarchar(20))	

The symbol  represents the field is the primary key.

3.1.17 AWARDINFORMATION Interface Fields

Users will be able to see the overall award information, such as creation of award information, number of recipients, total award and disbursement amount.

Field Name	Description	Data Type	Remarks
Action	C – Create; U – Update; D - Delete	String(\$nvarchar(1))	
Sourceld	Record unique Id	String(\$nvarchar(100))	A unique Id automatically generated by the database to ensure unique identification

 AwardId	Award record id	String(\$nvarchar(100))	Format: uuid
AwardBatchId	Award batch Id	String(\$nvarchar(50))	Sample Value: g105-001-00001
AwardName	Award name	String(\$nvarchar(100))	Sample Value: MOE Bursary
NoOfRecipients	Number of recipients	Int	Sample Value: 1, 2, 10
BudgetBalanceAmount	Budget balance available for award	Number(2 decimal)	Sample Value: 6000.00
PendingAwardAmount	Pending approval of award amount	Number(2 decimal)	Sample Value: 2000.00
AvailableBalance	Available balance for disbursement	Number(2 decimal)	Sample Value: 4000.00
CreatedTime	Date of creation	Datetime	Sample Value: 2023-02-28T00:00:00.000
Createdby	Award information created by	String(\$nvarchar(50))	Sample Value: Finance Admin
ModifiedTime	Award information modification date	Datetime	Sample Value: 2023-02-28T00:00:00.000
Modifiedby	Award information modified by	String(\$nvarchar(50))	Sample Value: Admin
Status	Status of award batch details	String(\$nvarchar(50))	Value list: Draft, Pending approval, Pending disbursement, Completed

The symbol  represents the field is the primary key.

3.1.18 STUDENTAWARD Interface Fields

Users will be able to review the information, and the applicable bursary tier. Upon approve, the finance users (holding the disbursement officers' role) will be notified to disburse the bursaries amount.

Below are the data fields for student award:

Field Name	Description	Data Type	Remarks
Action	C – Create; U – Update; D - Delete	String(\$nvarchar(1))	
SourceId	Record unique Id	String(\$nvarchar(100))	A unique Id automatically generated by the database to ensure unique identification
 StudentId	UAM Common Id	String(\$nvarchar(11))	

 AwardApplicationNo	Award application number	String(\$nvarchar(100))	Sample Value: Award ID-00001
RecipientId	Award recipient Id	String(\$nvarchar(100))	Sample Value: Award ARC00000001
AwardName	Award name	String(\$nvarchar(100))	Sample Value: MOE Bursary
ApplicationType	Award application type	String(\$nvarchar(100))	Sample Value: Award, Bursary, Incentive
IsCompanySponsored	Award is company sponsored	String(\$nvarchar(1))	Value List: Y-Yes; N-No
UEN	Company UEN. Displayed when the award is company sponsored	String(\$nvarchar(10))	Sample Value: OR220312
IsInSuspendedList	Student is in suspended list or not	String(\$nvarchar(1))	Value List: Y-Yes; N-No
EnrollmentStatus	Student enrollment status when apply the award	String(\$nvarchar(20))	Sample Value: Enrolled
MaritalStatus	Define marital status of the applicant.	String(\$nvarchar(20))	Value List: Single, Married, Devoiced, Separated, Widowed.
GrossMonthlyIncome	Total amount of all members monthly income.	Number (2 decimal)	Sample Value: 10000.00
GrossAnnualBonus	Total amount of all members annual bonus and awards.	Number (2 decimal)	Sample Value: 10000.00
NoOfFamilyMember	Total number of family members.	Int	Sample Value: 5
GHI	Total gross monthly household Income [GHI] Computational formula: GH=Gross monthly+ Total bonus and AWS/12	Number (2 decimal)	Sample Value: 11166.67
PCI	Per Capita Income [PCI] Computational formula: PCI=GH/Total number of family member	Number (2 decimal)	Sample Value: 2033.33
Remarks	Award remarks by CET admin	String(\$nvarchar(max))	
Amount	Award amount	Number (2 decimal)	Sample Value: 500.00
DisbursementMode	Award disbursement mode	String(\$nvarchar(100))	Sample Value: Cash
DisbursementAmount	Award disbursement amount	Number (2 decimal)	Sample Value: 250.00
DisbursementDate	Award disbursement date	Datetime	Sample value: 2023-02-28T00:00:00.000
ReversalAmount	Award reversal amount	Number (2 decimal)	Sample Value: 250.00

ReversalReason	Award reversal reason	String(\$nvarchar(max))	
AppliedTime	Award applied time	Datetime	Sample value: 2023-01-28T01:10:20.300
AwardStatus	Award status	String(\$nvarchar(100))	Sample Value: Pending disbursement, Application successful, Disbursed

The symbol  represents the field is the primary key.

3.1.19 STUDENTFAMILYMEMBER Interface Fields

The student family member form is used for applicants to create an award application.

Below are the data fields for student family member:

Field Name	Description	Data Type	Remarks
Action	C – Create; U – Update; D - Delete	String(\$nvarchar(1))	
SourceId	Record unique Id	String(\$nvarchar(100))	A unique Id automatically generated by the database to ensure unique identification
 FamilyMemberId	Student family member record id	String(\$nvarchar(100))	Format: uuid
StudentId	UAM Common Id	String(\$nvarchar(11))	
RelationshipToApplicant	Student family member relationship to applicant	String(\$nvarchar(20))	Sample Value: Self, Father, Mother
Name	Student family member name	String(\$nvarchar(100))	Sample Value: Tom
NRIC/FIN	Student family member NRIC/FIN number	String(\$nvarchar(20))	Sample Value: G1234567P
Age	Student family member age	Int	
Occupation	Student family member occupation	String(\$nvarchar(100))	Sample Value: Teacher
MonthlyIncome	Student family member monthly income	Number (2 decimal)	Sample Value: 5000.00
Bonus	Student family member bonus and awards.	Number (2 decimal)	Sample Value: 4000.00

The symbol  represents the field is the primary key.

3.1.20 EXAMSCHEDULE Interface Fields

Assessment scheduling section is used for Admin to view and manage assessment scheduling in the CET portal. Only scheduling for standalone exam and remediation exam will be managed and displayed in this section.

Below are the data fields for exam schedule:

Field Name	Description	Data Type	Remarks
Action	C – Create; U – Update; D - Delete	String(\$nvarchar(1))	
Sourceld	Record unique Id	String(\$nvarchar(100))	A unique Id automatically generated by the database to ensure unique identification
 StudentExamId	Student exam record id	String(\$nvarchar(100))	Format: uuid
StudentId	UAM Common Id	String(\$nvarchar(11))	
CourseCode	Course code	String(\$nvarchar(50))	Sample Value: TP-SD-FQ0001
CourseIntakeNo	Course intake number	String(\$nvarchar(100))	Sample Value: TP-SD-FQ0001-220001
ModuleCode	Code of the module	String(\$nvarchar(50))	Sample Value: BM0001
SubComponentName	Module sub-component name	String(\$nvarchar(20))	Sample Value: Final exam
ScheduleName	Exam schedule name	String(\$nvarchar(20))	Sample Value: Schedule1
SeatNo	Exam seat number	String(\$nvarchar(20))	Sample Value: S01
ExamDate	Exam date	Datetime	Sample value: 2023-02-28T00:00:00.000
StartTime	Exam start time	String(\$hh:mm)	Sample Value: 16:00
EndTime	Exam end time	String(\$hh:mm)	Sample Value: 18:00
FacilityName	Exam facility name	String(\$nvarchar(100))	Sample Value: Facility A
TrainerId	Exam invigilator Id, workday Id	String(\$nvarchar(20))	Sample Value: R12345;R54321

The symbol  represents the field is the primary key.

3.1.21 EXAMRESULT Interface Fields

Exam result section is for approvers granted approval for the student's score/ grade.

Below are the data fields for exam result:

Field Name	Description	Data Type	Remarks
Action	C – Create; U – Update; D - Delete	String(\$nvarchar(1))	
SourceId	Record unique Id	String(\$nvarchar(100))	A unique Id automatically generated by the database to ensure unique identification
 StudentExamId	Student exam record id	String(\$nvarchar(100))	Format: uuid
StudentId	UAM Common Id	String(\$nvarchar(11))	
CourseCode	Course code	String(\$nvarchar(50))	Sample Value: TP-SD-FQ0001
CourseIntakeNo	Course intake number	String(\$nvarchar(100))	Sample Value: TP-SD-FQ0001-220001
ModuleCode	Code of the module	String(\$nvarchar(50))	Sample Value: BM0001
AssessmentName	Assessment name	String(\$nvarchar(max))	Sample Value: CW1;CW2;Final
AssessmentWeightage	Assessment weightage	String(\$nvarchar(max))	Sample Value: 20%;20%;60%
AssessmentScore	Assessment score	String(\$nvarchar(max))	Sample Value: 80.00;85.00;90.00
AssessmentBaseScore	Assessment base score	String(\$nvarchar(max))	Sample Value: 100.00;100.00;100.00
BaseScore	Module base score	Number(2 decimal)	Sample Value: 100.00
ModuleScore	Module score. Calculate by assessment weight and score	Number(2 decimal)	Sample Value: 87.00
GivenScore	Module given score by trainer	Number(2 decimal)	Sample Value: 89.00
ScorePercentage	Calculated by Module given score/ Module base score	String(\$nvarchar(20))	Sample Value: 89%
ModuleGrade	Module grade	String(\$nvarchar(2))	Sample Value: A
GivenGrade	Module given grade	String(\$nvarchar(2))	Sample Value: A
AttendanceRate	Module session attendance percentage	String(\$nvarchar(20))	Sample Value: 90%
AssessmentStatus	Module assessment status	String(\$nvarchar(20))	Sample Value: DC(under investigation)

DetailStatus	Module assessment details status	String(\$nvarchar(20))	Value List: Unsubmitted, Pending approval, Confirmed, Reviewed
ModifiedOn	Exam record modified on	Datetime	Sample value: 2022-02-28T17:12:11.112
ModifiedBy	Exam record modified by	String(\$nvarchar(100))	Sample Value: Eric Zheng

The symbol  represents the field is the primary key.

3.2 Sample Files

The following sample CSV files are provided by AP:



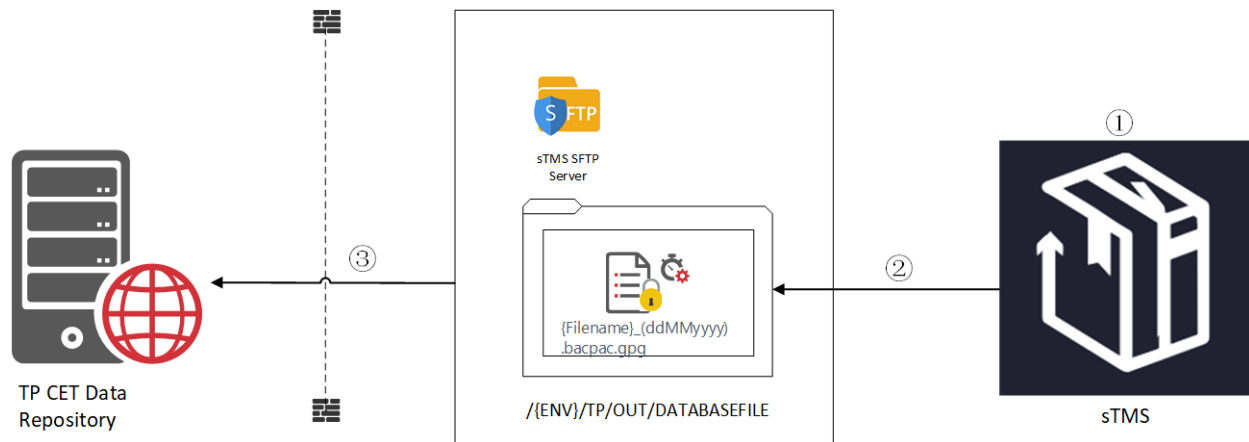
TP CET Data Repository Sample Files.zip

4. Database File Interface Design

4.1 Interface Diagram

There are 3 roles involved in the interface diagram.

- **TP CET Data Repository:** Temasek Polytechnic CET data repository.
- **sTMS SFTP Server:** A SFTP server hosted in Azure that stores encrypted files used for data exchange.
- **sTMS:** shared Training Management System.



4.2 Interface Design Summary

Interface File Name	{Filename}_(ddMMyyyy).bacpac.gpg
Data Source System	shared Training Management System
Data Destination System	Temasek Polytechnic CET Data Repository
Method	SFTP
File Path	/{ENV}/TP/OUT/DATABASEFILE
Encryption Method	PGP
Frequency	Weekly Sync
Data syncing mode	Full back up data. The backup file type is bacpac.
Timing	The timer job trigger timing is configurable, can be determine before system go live.

Folder Retention Policy	Files under the /{ENV}/TP/OUT/DATABASEFILE are retained for a maximum of 30 days in the sTMS SFTP server, after this period the files will be automatically deleted. TP also can delete these files after pulling out
Possible Exception and Exception Handling	File loss or transfer failure due to sTMS internal exception. E.g., Timer service down, etc. Exception handling steps: • sTMS support team shall be informed by email of the transfer failure. • AP team checks the relevant logs to troubleshoot the cause of the failure. • If the issue is due to the transfer files could not be found in the SFTP folder. ○ AP redeposits the files in the SFTP server.

4.3 Business Rule

Database file exporting rule

The TP full data in sTMS will be export to .bacpac file on 1 AM every Sunday morning(configurable in sTMS). The tables of this database file contain but not limited to the following:

- Course,
- Application,
- Enrolment,
- Timetabling,
- Assessment,
- Graduation,
- Student Finance,
- Master Data.

Filter condition:

- Institution = TP.

5. Assumption and Constraints

5.1 Assumption

No.	Assumption
1	sTMS is flexible to change the interface design of the tables if issue during UAT upon confirmation or any changes in the final sTMS design that impact with this interface.

5.2 Constraints

No.	Constraints
1	If CSV file interface field has multiple values (E.g., Component, Certificate, Trainer, etc.), combine strings and separate them with ";"

6. Appendix

6.1 Payment Collection Status

The payment status pertaining to each payment mode is as follows:

Payment Mode	Payment status	Definition
SFC	Pending	The claim submitted is pending approval
	Approved	The claim submitted has been processed successfully
	Rejected	The claim submitted has been rejected. Applicant may proceed back to payment summary to make payment. Both in-system notification and email will be sent to the applicant.
	Cancelled	The claim submitted has been cancelled by training provider or applicant
	Refunded	The SFC amount has been refunded to SSG. (To be further discussed in stage 3)
	Disbursed (Final status provided by SSG)	The SFC Claim has been paid out to training provider
	Paid (Finance users will manually update SFC status to "Paid" once IHL received the amount from SSG).	The SFC claim has been received successful by training provider (System will generate receipt upon SFC payment status = Paid)
PSEA (adhoc & SO)	Form received	The form has been submitted successfully
	Form verified & to be submitted to MOE	The form submitted has been approved by training provider and to be submitted to MOE for approval
	Approved by MOE	The form submitted to MOE has been approved
	Rejected by MOE	The form submitted to MOE has been rejected. Applicant may proceed back to payment summary to make payment Both in-system notification and email will be sent to the applicant.

Payment Mode	Payment status	Definition
	Pending resubmission	The form submitted is rejected. Applicant is required to resubmit the form or select another payment mode in payment summary to make payment
	Cancelled	The form submitted has been cancelled by training provider
	Paid	The PSEA amount has been paid to training provider
GIRO	Form received	The form has been submitted successfully
	Form verified	The form submitted has been approved by training provider and to be submitted to bank for approval
	Approved by bank	The form submitted to bank has been approved
	Rejected by bank	The form submitted to bank has been rejected. Applicant may proceed back to payment summary to make payment Both in-system notification and email will be sent to the applicant.
	Pending resubmission	The form submitted is rejected. Applicant is required to resubmit the form for processing or select another payment mode in payment summary to make payment
	Cancelled	The form submitted has been cancelled by training provider
	Paid	The claim has been paid out
Others	Paid	The payable amount has been paid successfully
	Unpaid	The payable amount has not settled
	Draft	Status available for manual creation of payment collection only
	Expired	For Pay-upon-application self-sponsored scenario only, upon payable amount is not settled within payment due date, the payment status will change to "expired"
	Cancelled	The payment is cancelled by training provider